Ogbonna Arinze

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PROFESSIONAL SUMMARY

Software engineer with 8 years of professional experience building user-facing products across AI, healthcare, fintech, civic tech, and enterprise software. Strong track record shipping fullstack and frontend features with React, Next.js, TypeScript, NodeJS, and modern web tooling, while collaborating closely with product managers and designers to turn complex systems into clear, usable interfaces. Experienced in end-to-end feature ownership, platform UI work, API-backed product development, CI/CD-driven quality, and AI-adjacent applications including production RAG systems and enterprise-facing conversational products.

CORE COMPETENCIES

- **Frontend Product Engineering:** React, Next.js, Vue.js, TypeScript, reusable UI systems, responsive layouts, and polished workflow-driven user interfaces.
- **Fullstack Delivery:** NodeJS REST APIs, NestJS services, authentication flows, API integration, and end-to-end feature ownership across frontend and backend systems.
- **AI Product Experience:** RAG systems, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search, streaming interfaces, and AI-powered user experiences.
- **State, Data, and UX:** Redux, Redux Toolkit, React Query, Axios, RxJS, Storybook, design systems, and data-heavy application workflows.
- **Quality and Performance:** Playwright, Jest, Enzyme, GitHub Actions, Lighthouse, React Profiler, code splitting, bundle analysis, and clean-code oriented delivery.
- **Infrastructure and Collaboration:** AWS, Heroku, Docker, monorepo workflows, CI/CD pipelines, agile delivery, cross-functional teamwork, and mentorship.

PROFESSIONAL EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main product for connecting and transferring clinical trial site data to various upstream EHR systems, migrating from a Vue.js web application to a Chrome extension in React.
- Maintained and extended the existing Vue.js codebase during the migration period, fixing bugs and shipping features in parallel with the React rebuild.
- Co-developed a full-stack Next.js chat widget powered by a Retrieval-Augmented Generation system using OpenAI GPT models, LangChain, Vercel AI SDK, and Chroma vector database, enabling users to connect their data sources, embed the widget on external sites, and query their data in real time via a conversational interface.
- Implemented authentication and authorization workflows on the frontend using JWT-based flows, securely handling access and refresh tokens with HTTP-only cookies, protected routes, and permission-based UI rendering in collaboration with backend engineers.
- Introduced Redux for state management and React Query for request caching, optimizing the build pipeline and refactoring legacy React code from class components to functional components.
- Optimized application performance with lazy loading, code splitting, image optimization, responsive layouts, memoization, and efficient asset delivery via CDN, using Lighthouse and Chrome DevTools to identify and fix bottlenecks.
- Increased stakeholder confidence by writing Playwright end-to-end tests in the CI/CD pipeline, publishing Allure reports on GitHub Pages, and sending Slack notifications with links to generated test results on every push.
- Architected a reusable design system in React and Radix UI that improved UI consistency, accessibility, and developer efficiency across products, while mentoring two junior developers.

OurNigeria — ournigeria.ng

Senior Engineer — *Feb 2026 - Present*

AI-powered platform for Nigerian budget analysis and corruption tracking, processing 700+ budget documents and 708K+ vector embeddings across 37 states.

- Architected and built the entire platform as an Nx monorepo managing 6 applications: NestJS API backend, Next.js citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed and implemented a multi-agent AI system using the Mastra framework with 5 specialized agents for intent classification and domain-specific RAG-powered analysis.
- Built a hybrid RAG pipeline combining pgvector similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion with Cohere reranking and Voyage AI embeddings across 4 vector indexes.
- Built the Next.js citizen-facing frontend with React 19, Tailwind CSS v4, and Recharts, featuring 22 interactive chart types, responsive mobile layouts, and AI-generated data presentation.
- Configured CI/CD with GitHub Actions, Playwright end-to-end tests across 4 projects, and Docker Compose production deployment with health checks, memory limits, and graceful shutdown behavior.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive user interface using React, achieving a 25% boost in user engagement while collaborating with software architects to create a scalable frontend architecture and accelerate development by 30%.

- Enhanced the development workflow by introducing automated testing frameworks, including Jest and Enzyme, which led to a 20% decrease in bug reports.
- Utilized Storybook for documenting React components, streamlining change management and ensuring clear communication of development processes while supporting agile user-story planning and estimation.
- Drove a 40% improvement in React application performance by optimizing caching mechanisms and significantly boosting speed and user experience.
- Optimized frontend performance and load efficiency by identifying and resolving render bottlenecks using React Profiler and Lighthouse Audits, reducing bundle size by 30-40% through rollup-plugin-visualizer and webpack-bundle-analyzer, which led to a 35% faster load time.
- Established engineering standards for code quality, testing, and deployment using ESLint, Prettier, Husky, Lint-staged, and GitHub Actions, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers and guided frontend best practices, fostering a culture of high-quality, maintainable code and consistent design across projects.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Translated business requirements into efficient database models and schemas, collaborating closely with the business team to ensure technical solutions met business needs.
- Developed NodeJS RESTful APIs to power new features, focusing on testable, maintainable, and well-documented code.
- Transformed user journeys and Figma designs into reusable React components, ensuring a consistent, intuitive user interface.
- Engineered a CI/CD pipeline for automated deployment of APIs to testing environments on Heroku and production servers on AWS, while guiding two interns through frontend and backend projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Played a pivotal role in a team of 3 frontend engineers to build a high-volume bank Settlement Solution, significantly enhancing transaction processing capabilities to handle thousands to millions of transactions daily, boosting transaction settlement rates by 80%.
- Spearheaded the development of critical modules for the Settlement Solution, including MasterCard, VISA, Express-cash, and E-commerce modules, using React with Redux for state management, RxJS for data streaming, and Axios for asynchronous API requests.
- Developed a real-time account reconciliation solution, boosting the reconciliation of daily transactions by 60% through innovative transaction processing and data management strategies.
- Enhanced system performance and user experience by adopting the native browser EventSource API for real-time data streaming and built dynamic data visualization dashboards using React, react-chartjs, Redux Toolkit, and Ant Design UI.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed MongoDB database schemas and implemented them in NodeJS for multiple applications.
- Worked with a team of 5 to develop SureEdu, a school management portal for school owners to manage staff, students, and guardians.
- Responsible for server-side web application logic as well as integration of the frontend, including use-case definition, schema design, documentation, and testing of functions, triggers, and API endpoints.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science, 4.1 CGPA *Sep 2019*